

Do (Not) Follow the White Rabbit: Challenging the Myth of Harmless Open Redirection

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Author not stated, NDSS Symposium.

- Published: date not stated
- Original: <<https://www.ndss-symposium.org/ndss-paper/do-not-follow-the-white-rabbit-challenging-the-myth-of-harmless-open-redirection/>>
- Preserved from: <https://www.ndss-symposium.org/ndss-paper/do-not-follow-the-white-rabbit-challenging-the-myth-of-harmless-open-redirection/> (live) on 2026-08-08
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Open redirects are one of the oldest threats to web applications, allowing attackers to reroute users to malicious websites by exploiting a web application's redirection mechanism. The recent shift towards client-side task offloading has introduced JavaScript-based redirections, formerly handled server-side, thereby posing additional security risks to open redirections. In this paper, we re-assess the significance of open redirect vulnerabilities by focusing on client-side redirections, which despite their importance, have been largely understudied by the community due to open redirect's long-standing low impact. To address this gap, we introduce a static-dynamic system, STORK, designed to extract vulnerability indicators for open redirects. Applying STORK to the Tranco top 10K sites, we conduct a large-scale measurement, uncovering 20.8K open redirect vulnerabilities across 623 sites and compiling a catalog of 184 vulnerability indicators. Afterwards, we use our indicators to mine vulnerabilities from snapshots of live webpages, Google search and Internet Archive, identifying additionally 326 vulnerable sites, including Google WebLight and DoubleClick. Then, we explore the extent to which their exploitation can lead to more critical threats, quantifying the impact of client-side open redirections in the wild. Our study finds that over 11.5% of the open redirect vulnerabilities across 38% of the affected sites could be escalated to XSS, CSRF and information leakage, including popular sites like Adobe, WebNovel, TP-Link, and UDN, which is alarming. Finally, we review and evaluate the adoption of mitigation techniques against open redirections.

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